



UNIVERSAL COFFEE MACHINES
Architect Workshop Scenario

CONFIDENTIAL



Project Overview

Universal Coffee Machines (UCM) Ltd. produces coffee machines mainly for the luxury consumer market, with high-quality machines that require installation in customers' homes. UCM's headquarters is in Australia, with support offices in the US, Singapore, UK, and Italy. They design the machines in Australia, and they have a factory in China that manufactures the parts. Depending on the model, they assemble them in either China or Australia. Post manufacturing, UCM leverages a network of authorized resellers, with staff specifically trained to provide a personalized shopping experience. It also has a network of authorized technicians who help install and provide after-sale, in- and out-of-warranty servicing.

Traditionally, UCM focuses only on their resellers and technicians instead of their customers, with the exception of warranty registrations and covering costs of in-warranty services. UCM is embarking on a transformation. They wish to build a closer customer connection to strengthen customer loyalty. A key part of the transformation is to enable "connected capability": UCM's updated line of coffee machines can now send usage pattern data and warning / error messages of failing machine parts back to UCM. With this, UCM wants to:

1. Analyze usage pattern information, allowing UCM to build better coffee machines.
2. Streamline after-sale services by monitoring any machine issues, and send out technicians or "repair packs" to customers, potentially before they realize there were any faults.
3. Encourage customers to connect their machines back to UCM by providing them with timely maintenance notifications in return.

System Landscape

Currently, UCM uses a number of systems to run their business:

1. A public website which provides product information, phone numbers, and enquiry forms to customers, resellers, and technicians in all the major markets.
2. A product design application suite that contains all coffee machines' design specifications, user manuals, service manuals, and FAQs. It stores data in an FTP-based file system.



3. Two separate ERP systems, one in the China and one in Australia.
 - a. The Australian office's ERP system manages sales, manufacturing (in sending parts requests to the Chinese factory, and assembly operations in Australia), installation, and after-sale services. The other support offices are currently using Australia's ERP system.
 - b. The Chinese office's ERP system purely focuses on manufacturing (manufacturing parts and assembly operations in China).
 - c. The ERP systems will remain separate, and are likely to be running different versions of the vendor's SOAP API at any given time.
4. A custom data collection application, currently hosted on-premise in the Australian office, to collect usage pattern data and warning / error messages from connected coffee machines.
 - a. The data can then be queried via a REST API.
 - b. The application can push messages based on specific events (e.g., error messages or specific events, such as the 100th cup of coffee) to URL end-points in JSON format.
5. The Australian and European offices use one Windows Active Directory for management of internal users across all systems; China uses its own Linux LDAP server for users to access the ERP.

UCM wishes to:

1. Decommission the sales, installation, and after-sale service operations of the Australian office's ERP system, and replace them with Salesforce.com.
2. Provide self-serve access for resellers, technicians, and customers to the relevant processes.
3. Integrate the data collection application for it to perform appropriately at scale globally; and enable automated services workflows to deal with warning / error messages or relevant customer notifications based on specific usage events.
4. Decommission the current website and migrate it into Salesforce.com to provide a seamless public site for access to the most up-to-date product and reseller information.
5. Integrate the product design and engineering application suite to eliminate error-prone manual entry and allow product information to be displayed in the public site, made available to existing resellers, technicians, and customers, and used by the relevant ERP systems for manufacturing operations.

Given the differences in the separate back-end systems between Australia and China, UCM is looking for guidance on the approach to the global Salesforce implementation.



Functional Requirements

The following are functional requirements for the proposed system.

The public website:

1. The general public can browse the website for product information and search for the nearest reseller.
2. Prospect organizations can apply on the site to become an authorized reseller or technician for UCM's coffee machines. They fill in a form which is forwarded to UCM's Partner Support team.

UCM partner portal:

1. An application is reviewed by UCM Partner Support, who ensures they are qualified to work with UCM. Once qualified, the organization is authorized as a reseller or technician, or both.
2. A key contact in the organization will be given access to the UCM partner portal. This allows the key contact to access relevant portal functions on their organization's behalf.
3. The key contact can add more reseller staff / technicians under the same organization.

Sales and Installation:

1. When the customer arrives at the reseller's store, the reseller checks to see if the customer's record already exists before creating a new record. Resellers can see all customers' contact details.
2. The staff then attends to the customer, and at the end log an interaction or record a sale.
3. When the reseller staff records the sale in the system, the order is forwarded to UCM Partner Support, who prepares the machine and records the machine serial number against the customer for warranty purposes.
4. The order is then assigned to an authorized technician, who picks the machine up from UCM's office, travels to the customer's home, installs the machine, asks the customer to allow them to enable the machine's connected capability, and finally connects to the coffee machine, allowing it to send data to UCM's data collection application.



“Connected capability,” data collection, and output:

1. Once connected, the coffee machines sends usage data to the data collection application managed by UCM. The messages represent every single action the machine has taken, as well as outcomes of periodic self-checks and any warnings / errors detected in the machine.
2. For any warning / error messages received, the data collection application can, in real time, pass through details of the machine’s serial number and the identified error into an endpoint.
3. Based on the type of message, UCM wants either a timely notification sent to the customer or a relevant service request being created and managed in Salesforce.

Service process:

1. Service requests can be raised by customers via self-serve access, generated by a machine's error or warning message, or other traditional channels (e.g., website queries, email, or phone).
2. When a service request is raised, it is reviewed by UCM Partner Support to determine the request severity, priority, and how to resolve the request. Once a request is reviewed, the resolution can be actioned in one of two approaches: self-repair or authorized technician visit.
3. For self-repair, a kit with the relevant parts and instructions are sent to the customer; the request remains open until the machine sends out a “pass” message by a self-check triggered by the installation of the repair parts. If after a certain time no such message is received by UCM, the request is escalated into an authorized technician visit.
4. For an authorized technician visit, the request is assigned to a technician, who is notified to check further details via self-serve access; he or she will contact the customer and organize a visit to repair the machine, and updates the request to “completed.”
5. The technician will raise the relevant invoice to UCM and record it against the same service request on completion. UCM absorbs the cost if the machine is under warranty; otherwise UCM passes on the invoice to the customer.
6. Customer can see all service requests that are raised and resolved on the machine during their ownership, but they can see the relevant invoice details only if it was an out-of-warranty repair and they need to pay the invoice.



Data Requirements

1. There are over 1M customer records and 1.2M machine records in the ERP, but currently there are only 300,000 customers globally, with 400,000 active machines that are under warranty.
2. UCM expects that the number of customers will grow by 90,000 per year, and they will sell up to 120,000 machines per year.
3. UCM expects around 850k enquiries by prospective customers, resellers, or technicians per year.
4. UCM has about 10,000 authorized resellers globally, with an average of 5 employees each, and about 5,000 authorized technicians globally, half of which are individual contractor. The other half are service companies with an average of 3 authorized technicians.
5. UCM expects that the messages generated from the machines' connected capability will result in 500,000 notifications and 150,000 service requests each year.
6. Internally, UCM has the following staff:
 - a. UCM Partner Support: About 100 users; around half the users in sales, and the rest in service. One-third of the staff is in Australia, and the rest are spread out globally. They provide support to authorized resellers and technicians.
 - b. UCM Design & Manufacturing: 50 in Australia and 20 in China, who focus on the design, manufacturing, and assembly of the coffee machines.
 - c. UCM Management: About 20 users, with visibility into all operations within UCM.
7. From a legacy data standpoint, the following data will need to be made available to users in the new solution for go-live:
 - a. Contact details of all customers with active machines that are still under warranty.
 - b. All active machines' product model and serial number details, warranty details, and ownership details of those machines linking back to the customer records.
 - c. The link between all those active machines and the resellers who sold them.
 - d. All 100,000 previously raised service requests linked back to the relevant existing machines and customers who raised them.
 - e. The link between those service requests and the technicians who worked on them.
 - f. The invoices for all previously raised service requests.



8. The current ERP System has data quality issues, such as duplicate data, incorrect addresses, and incorrect email addresses.
9. Deployment can occur over the weekend, but legacy data migration from existing systems will take over a week.

Reporting Requirements

1. For customer, machine, ownership, and warranty information, both the legacy data AND migrated data (e.g., for customer records, both the 300,000 current customer records and the remainder of the 1M records that do not need to be made available for the new solution) need to be made available for reporting, together with any new data created after solution go-live.
2. For service requests and invoices, and resellers / technicians' relationships with the relevant sales / service requests, any new data post go-live needs to be reportable in the new solution; however, legacy data of those records should be made available to all users for a complete view of the customers' and machines' prior history.